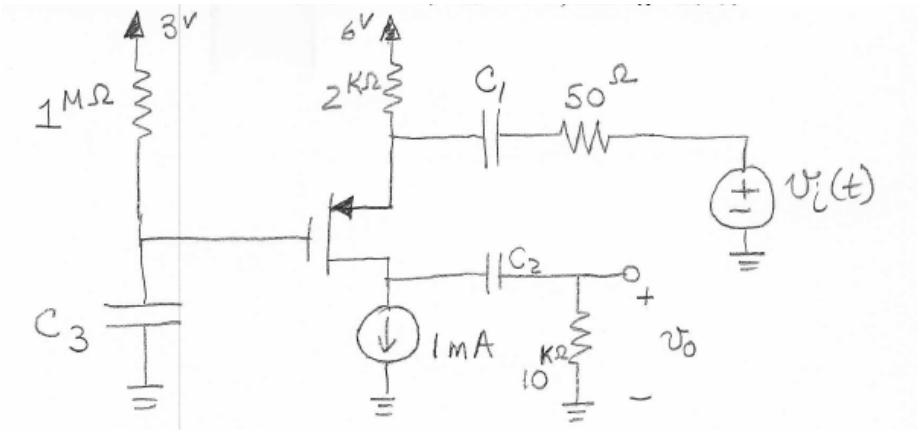
**ECE 3143** **HOMEWORK #6** **Due 11/11/2026**

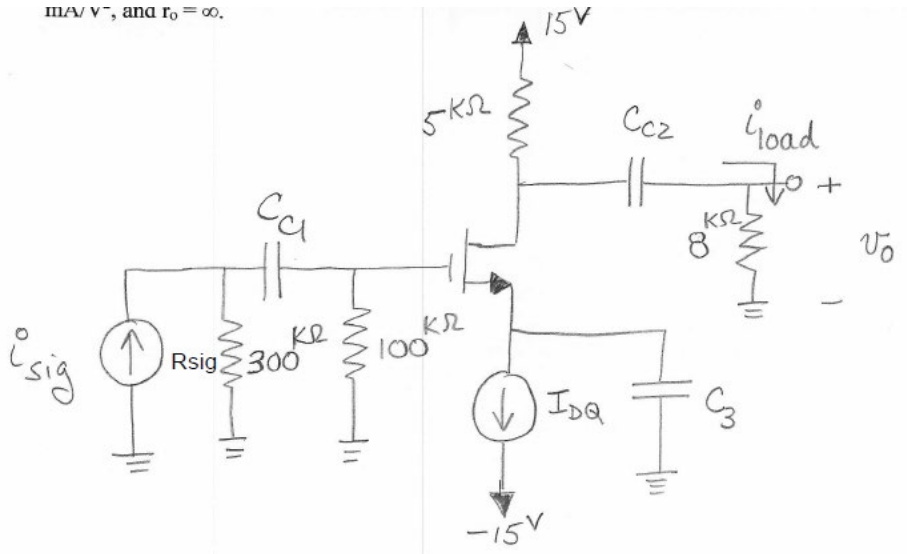
To Be Graded:
**Chapter 7:** Problems 33, 59, 66

**Note:**. For problem 59, you are using the box/2-port model for the amplifier.

**Problem 7.A-1** In the circuit below, ½k = 4 mA/V$^2$, $V_t$ = -0.5 V, $V_{GS}$ = -1V and $\lambda$ = 0.02 V$^{-1}$. (a) Determine $g_m$ and $r_o$ for the FET. (b) Draw the small-signal model of the amplifier circuit. (c) Determine $A_v$, and $R_i$ for the amplifier, (d) What is the largest value of $v_i(t)$ for which the small-signal condition is satisfied by at least a factor of 5?




**Problem 7.A-2** For the amplifier circuit below, the FET parameters are $V_t$ = 1 V, ½k = 0.25 mA/V$^2$, and $r_o = \infty$. (a) Which amplifier configuration is this? (b) What value of $I_{DQ}$ will make the amplifier have a voltage gain $A_v$ = -8 V/V? (c) If $I_{DQ}$ = 1 mA, use either of the box models discussed in class to determine the overall current gain $G_i = i_{load}/i_{sig}$.




**continued on back/next page**

**Problem 8.A-3** For this problem, use the box models for the amplifiers in problems 8.A-1 and 8.A-2 only, do not include any of the signal sources or signal and load resistors form those problems. If you put the two amplifiers in series (one after the other), which one would you put first, which would you put second, to get the largest overall gain $G_v$?

Practice:
**Chapter 7:** Problems 29, 31, 118(a, b, c),